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ORIGINAL ARTICLE

Performance evaluations of large language models for customer service

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Abstract
Recent advances in large language models (LLMs) show broad promise for a variety of natural language processing (NLP) tasks. There is growing interest in LLMs for domain-specific applications, such as in the telecommunications customer service domain. However, most of the benchmarks are for open-source LLMs, lacking effective exploration of real scenarios. To explore the performance of LLMs for customer service, we propose the first Telecommunications Customer Service Evaluation Benchmark (TeleEval-CS). In our work, we simulate the customer service pre-call, in-call, and post-call using 8.1k examples of 15 subtasks containing 21 datasets. We build 90k domain-specific multi-tasking instruction samples and fine-tune three types of LLMs including basic NLP, knowledge Q&A, and multi-round dialogue to realize industry knowledge injection. We conduct a comprehensive evaluation in multiple scenarios on 34 open-source, 5 closed-source, and 4 fine-tuned LLMs with zero-shot and few-shot approaches on TeleEval-CS. Experimental results show that the open-source LLMs can also perform better than the closed-source LLMs. The performance of the fine-tuned LLM depends on the quality of the fine-tuning dataset rather than its size, and fine-tuning has great potential in customer service scenarios. We provide our data and code at https://github.com/zsjslab/TeleEval-CS.

Keywords Benchmark · Large language model · Telecommunications customer service domain

1 Introduction

With the proposal of GPT-4 [1] and the popularity of the application ChatGPT,[1] large language models (LLMs) (e.g. LLaMA [2, 3], GLM [4, 5], Baichuan [6], etc.) have achieved remarkable results and garnered wide attention across various NLP tasks. Since these LLMs do not require labeled data and can achieve significant results using zero-shot or few-shot approaches, how effective LLMs are in a wider range of scenarios and industries, such as legal [7, 8], medical [9, 10], financial [11, 12], etc., has attracted active exploration, and telecommunications customer service is no exception.

With the fast development of LLMs, it is critical to tell users which LLM they should use for their own tasks. Thus we need to know the capability of different LLMs. Numerous benchmarks have been proposed for this purpose, and different benchmarks focus on different aspects. For example, MMLU [13], Bigbench [14], CMMLU [15], etc. are mainly dedicated to mining the NLP capability in general-purpose domains. CEval [16], AGIEval [17], etc. concern the performance on real human tasks. There are also a number of evaluation benchmarks for specific domains, such as legal benchmark (LAiW [18]), medical benchmark (PromptCBLUE [19]), financial benchmark (PIXIU [20]), and so on. One important application area for LLMs is customer service. However, it is difficult for these benchmarks to truly understand the performance of LLMs in real telecommunications customer service scenarios [21, 22].

In the realm of customer service in China, the three major operators have introduced prominent LLMs designed for

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[1] https://openai.com/blog/chatgpt.

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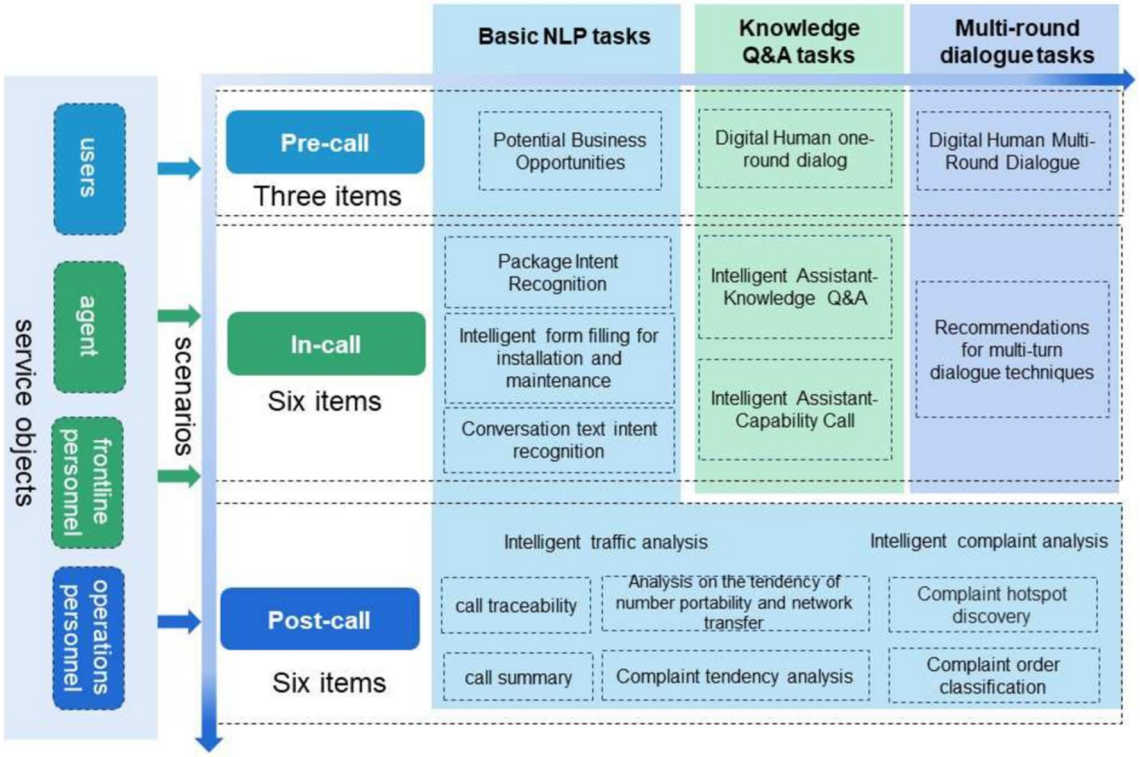


Fig. 1 Overview of the framework in telecommunications customer service domain

specific vertical fields. China Telecom introduced Telechat [23], China Unicom presented the "Honghu Graphic LLM",[2] and China Mobile launched the "Jiutian Customer Service LLM".[3] However, these LLMs are mostly closed-source and primarily focus on assessing conversational abilities. They lack a comprehensive and systematic evaluation of multi-tasking throughout the entire customer service process, including pre-call, in-call, and post-call scenarios.

Furthermore, the uniqueness of the telecommunications customer service domain makes it difficult to reasonably assess through existing LLMs and measurement benchmarks. Firstly, as shown in Fig 1, this field has complex scenarios and diverse tasks, covering three major categories pre-call, in-call, and post-call. Each of these categories contains various tasks, such as multi-cascade intent recognition, information extraction, and multi-round dialogue tasks. Secondly, limited by dialects and speech-to-text technology, the quality of customer service texts is poor. Finally, the dialogue data contains sensitive information and needs to be desensitized before use. In order to truly understand the performance of LLMs on a series of real telecommunications customer service scenarios, it is essential to build a systematic and comprehensive benchmark for the operational customer service domain.

To the best of our ability, we propose a comprehensive benchmark for telecommunications customer service, addressing the need for a systematic evaluation. Firstly, we categorize customer service scenarios into 15 subtasks based on the three phases of customer service: pre-call, in-call, and post-call, and define three types of capabilities: basic NLP, knowledge Q&A (Question and Answering), and multi-round dialogue. Secondly, we simulated an evaluation dataset of more than 8.1k examples based on LLMs in accordance with the telecommunications customer service task characteristics. Thirdly, we collect 90k instruction samples to fine-tune domain-specific LLMs. Finally, we evaluate the performance of various LLMs in customer service domain scenarios.

The main contributions of our work can be summarized as follows:

- The Telecommunications Customer Service Evaluation Benchmark (TeleEval-CS) benchmark proposed in this paper is the first benchmark that can be used to comprehensively assess the pre-call, in-call, and post-call capabilities of different LLMs, and it is an important catalyst to promote the high-quality development of the telecommunications customer service domain.

[2] https://baike.baidu.com/item/鸿鹄.

[3] https://baike.baidu.com/item/九天客服大模型.

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- We combed over 8.1k assessment examples of 21 subtasks and over 90k fine-tuning data to train telecommunications customer service multi-tasks LLMs, including customer service basic NLP tasks, knowledge Q&A tasks, and multi-round dialogue tasks. In addition, we also use all the instruction data to fine-tune the TeleFtLLM-7B (full) and conduct ablation study.
- We have evaluated three categories of LLMs, including 34 open-source, 5 closed-source, and 4 fine-tuned LLMs in customer service scenarios. The results provide an important reference value for guiding technological innovation and product iteration, and promote iterative upgrades of large model applications in the customer service domain. What's more, it provides feasible suggestions for intelligent upgrading of telecom enterprises.

We summarize the key findings as follows:

- Closed-source LLMs are not irreplaceable, and open-source LLM Qwen-1.5-32B performs the best over most of the tasks, while Qwen-1.5-14B-Chat achieves comparable performance to that of the open-source LLMs.
- Most LLMs perform better in 5-shot NLP tasks than in 0-shot NLP tasks.
- Furthermore, fine-tuning can indeed effectively improve the performance of some tasks, but it is closely related to the quality rather than the size of the instruction data.

2 Related work

2.1 General LLMs and benchmarks

Recent advancements in LLMs, such as GPT-4 [1], have demonstrated outstanding capabilities in handling different issues in general and professional domains. Many researchers have begun attempting to replicate the GPT series to develop open-source foundational LLMs such as LLaMA [2, 3], Bloom [24], Qwen [25], etc. The success of LLMs largely depends on the benchmark datasets used to assess their proficiency in specific domains. These datasets also play a crucial role in determining the architecture design of LLMs and guiding the pre-training and fine-tuning processes for these specialized domains.

Traditional NLP benchmarks like GLUE [26], SuperGLUE [27] and CLUE [28] can be regarded as comprehensive evaluation standards formed by various NLP tasks. They primarily assess model capabilities across single and multiple NLP-specific tasks, such as sentiment analysis, reading comprehension, etc. While these benchmarks were widely adopted during the BERT [29] era, they are not well-suited for the recent LLMs that exhibit powerful capabilities in complex reasoning and problem-solving tasks. To evaluate the advantages and limitations of LLMs, some new benchmarks have been proposed to evaluate the broader knowledge and advanced abilities of LLM, mainly transitioning from traditional NLP tasks to simulate human exams, such as M3KE [30], AGIEval [17] and CEval [16]. Further, some researchers suggest that natural language generation (NLG) with multi-round dialogue should be evaluated as a core competency [13, 31]. LLM-as-a-judge [31] constructed two benchmarks: MT-bench, a multi-round question set, and Chatbot Arena, a battle platform to verify the agreement between LLMs and human preferences. Although general LLM evaluation benchmarks have been developed comprehensively, due to the uniqueness of different fields, general benchmarks cannot be directly applied to the evaluation of specific fields.

2.2 Telecommunications LLMs and benchmarks

LLMs have shown remarkable performance in many fields, such as legal [7, 8], medical [9, 10, 32, 33], financial [11, 12], etc. However, their utilization in the telecommunications industry is still relatively limited. Currently, in the telecommunications customer service domain, the three major operators have successively introduced self-developed vertical domain LLMs. China Telecom has released Telechat [23] LLM, covering three directions: a model empowering data center, intelligent customer service, and intelligent government affairs [34]. China Unicom has launched the "Honghu Graphic LLM", the first LLM tailored for the operator value-added services. It enables various functions such as text-to-image generation [35], video editing, and image-to-image generation. China Mobile has created the "Jiutian Customer Service LLM" based on the massive service data, business knowledge, and service experience accumulated over many years in the field of customer service [36, 37].

Although these LLMs are focused on the operator industry, they are mostly closed-source. Moreover, only a brief evaluation of these LLMs has been carried out in the dialogue capability, and an objective, comprehensive, multi-task, and multi-scenario scientific systematic evaluation system is lacking [38]. As the first public benchmark dataset for evaluating LLM knowledge in the telecommunications field, TeleGnA [39] covers 10,000 questions and answers across various disciplines in the telecommunications field. However, it does not use data from real-life scenarios of telecommunications customer service, and lacks scientific system evaluation of multi-tasking across the entire pre-call, in-call, and post-call scenarios of customer service. Therefore, there is an urgent need to develop an assessment suite for LLM in telecommunication customer service to meet the growing demand in this field. To confront this, our TeleEval-CS combines real telecommunication customer service scenarios, systematically organizing the evaluation dataset from three

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perspectives: pre-call, in-call, and post-call. In particular, we consider the specific task form and design three levels: basic NLP tasks, knowledge Q&A tasks, and multi-round dialogue tasks to fully evaluate the capabilities of LLMs [40].

3 Benchmark construction

Collaborating with experts in the fields of telecommunications customer service and AI (Artificial Intelligence), we have categorized the capabilities of customer service from the perspective of service timeline and feasibility, and established an expandable multi-scenario and multi-task evaluating system. As shown in Fig. 1, in terms of scenarios, we divide the customer service domain into pre-call, in-call, and post-call phases. Regarding capabilities, we categorize the customer service tasks of LLMs into three levels, ranging from easy to difficult: basic NLP, knowledge Q&A, and multi-round dialogue, with a total of 15 subtasks.

3.1 Pre-call scenarios

In the pre-call scenario, the essential capabilities involve providing proactive service to users before initiation by customer service agents. This aids in addressing common issues proactively and decreasing the number of agent-initiated calls.

- Basic NLP capability: Primarily by analyzing the previous call data to identify potential business opportunities that can be leveraged, such as mining tasks for potential business opportunities.
- Knowledge Q&A: Users will first access the digital assistant when they have some questions about broadband, traffic, etc. The digital assistant is capable of addressing basic inquiries primarily related to phone bills, traffic, etc., aiming to alleviate the workload of the agent.
- Multi-round Dialogue: Before users access the real agent, they will first access the voice navigation multi-round dialogue system to answer some simple questions from users.

3.2 In-call scenarios

In-call services mainly refer to the ability to provide various types of services when users have actual conversations with customer service. In-call services are used to help back-office agents improve service efficiency and reduce the processing time of each call.

- Basic NLP capability: Including two categories, the first is the intent recognition task, mainly to identify the user's intent on various users and customer service dialogue text, such as business processing to determine whether they want to apply for "broadband class (宽带类)" business or "points class (积分类)" business. These two terms are specific to the telecommunications field, where "broadband clas" refers to services related to broadband, and "points class" refers to services related to user points. The second category is information extraction tasks, mainly involving intelligent form filling, extracting key information in the conversation, and filling in fixed forms to reduce manual operations.
- Knowledge Q&A capability: It encompasses knowledge quizzes and capability calls. Knowledge quizzes involve providing users with answers to their questions by consulting pertinent knowledge base documents. Capability calls refer to the ability to respond when users seek information, such as traffic details and remaining package data, by invoking multiple backend interfaces.
- Multi-round Dialogue capability: The primary purpose is to generate targeted reply recommendations, aiding seat personnel with specific reply scripts during the dialogue process between users and customer service.

3.3 Post-call scenarios

Post-call service covers intelligent call analysis and intelligent complaint analysis. The former is used to examine common problems such as call sources, key information, and complaint tendencies, so as to identify centralized user needs, with an insight to make subsequent improvements and reduce call volume. The latter focuses on identifying complaint hotspots and types in customer service complaint work orders to reduce the number of complaints. The above purpose is to assist backend staff in analyzing call data and monitoring whether complaints are resolved to improve user satisfaction.

4 Datasets construction in customer service domain

To evaluate the mentioned capabilities and corresponding tasks, we integrated and constructed telecommunications customer service evaluation datasets and supervised fine-tuning (SFT) datasets, based on open-source data and private data. They both undergo a similar process, as shown in Fig. 2, which includes data desensitization, data cleaning, answer or label construction, and manual verification. First, to ensure user privacy and security, we identify and desensitize numerical texts through regular expressions. Then, we utilize Qwen-14B-Chat for text recognition, such as name, address, and other information, to replace special characters. Second, meaningless characters such as underlines and spaces from the desensitized text are removed. Third, we

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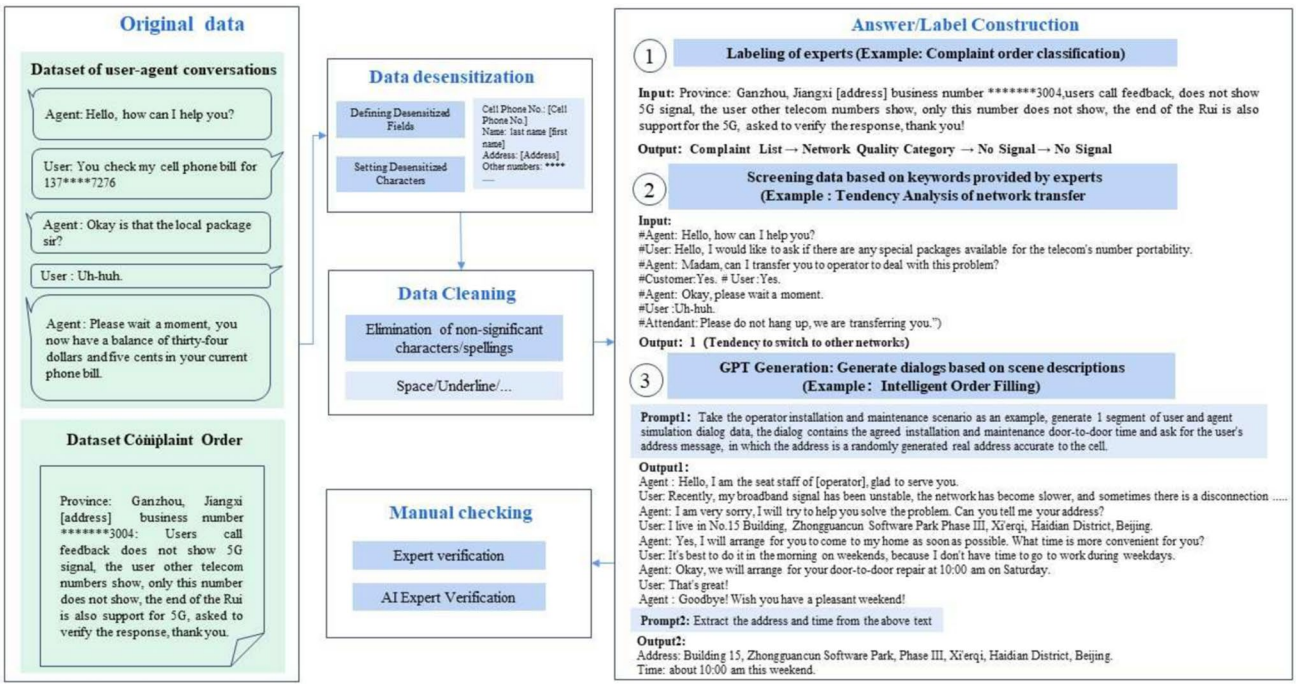


Fig. 2 Construction of Datasets for evaluation and fine-tuning two stages

design appropriate answers or labels for each task, including a) expert annotation, b) keyword filtering, and c) using LLMs to generate samples that meet requirements based on scene descriptions. Finally, all these datasets are verified by three business and NLP experts to ensure the reliability of the data. In order to enrich the data resources, it is necessary to improve the training effect and generalization ability of the model, and better meet the needs of various application scenarios. In addition to the telecommunications customer service evaluation datasets, open source data is also introduced, including CSDS [41] and MUSIED [42], and they have been processed with reference to the operator domain characteristics.

The following introduces the telecommunications customer service evaluation datasets and SFT datasets.

4.1 Evaluation datasets

We construct 8.1k examples of 21 datasets containing 15 subtasks for evaluating telecommunications customer service, including basic NLP, knowledge Q&A, and multi-round dialogue tasks. The open-source data includes CSDS and MUSIED. The CSDS dataset, which contains multiple rounds of dialogue, intent, and summary information, has been split into multiple task datasets, including CSDS-MRD, CSDS-IR, and CSDS-AE. The MUSIED dataset contains user comments and intent information, hence we construct it as the intent recognition dataset MUSIED-IR. In addition, we also constructed a private dataset using the method shown in Fig. 2. The specific statistical information is shown in Table 1.

4.2 Supervised fine-tuning datasets

By training with a small yet diverse domain dataset, the model can adapt to new tasks faster and improve performance. Compared to models without fine-tuning, fine-tuning large models performs better on new domain tasks.Therefore, it is meaningful to evaluate the performance of the model after fine-tuning in the operator field. We constructed an SFT dataset to further explore the potential of LLMs in the telecommunications industry. First, we collected as much open-source and private domain data as possible, totaling 300,000 texts. Previous researches [43, 44] has demonstrated that the quality of training data plays a crucial role in the performance of downstream tasks, poor-quality data can have adverse effects on the performance of LLMs. Similar to the process of the evaluation dataset, we constructed 90k high-quality examples from all collected data. As shown in Table 2, it is primarily composed of 3 task datasets, focusing on fine-tuning the LLM capabilities of basic NLP, knowledge Q&A, and multi-round dialogue in the customer service field.

We design several instruction templates for each task to build SFT datasets. For basic NLP tasks, approximately 5-10 instruction templates are manually created for each task. For knowledge Q&A tasks, the original form can satisfy the instruction fine-tuning. The multi-round dialogue consists

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Table 1 Overview of the evaluation dataset and metrics

Scenes	Dataset	Task	Major task type	Minor task type	Size	Metrics
Pre-call	PBO	Potential Business Opportunities	NLP task	Text classification	100	Accuracy, Instruction_follow
	DHMRD	Digital Human Multi-Round Dialogue	Multi-round dialogue task	Multi-round dialogue	20	Uni, GT, ELO
In-call	CTIR	Conversation Text Intent Recognition	NLP task	Text classification	225	Accuracy,Instruction_follow
	PIRC	Package Intent Recognition of Consumption	NLP task	Text classification	1013	Accuracy,Instruction_follow
	PIR16	Package Intent Recognition_16 classes	NLP task	Text classification	277	Accuracy,Instruction_follow
	PIR3	Package Intent Recognition_3 classes	NLP task	Text classification	310	Accuracy,Instruction_follow
	IAKQA	Intelligent Assistant-Knowledge Q&A	Knowledge Q&A task	Knowledge Q&A	54	RougeL
	IACC	Intelligent Assistant-Capability Call	Knowledge Q&A task	Capability call	20	RougeL
	IFF	Intelligent Form Filling for Installation and Maintenance	NLP task	Information extraction	100	F1_score
	CSDS-MRD	CSDS-Multi-Round Dialogue	Multi-round Dialogue task	Multi-round dialogue	100	Uni, GT, ELO
	CSDS-IR	CSDS-Intent Recognition	NLP task	Text classification	1000	Accuracy,Instruction_follow
	MUSIED-IR	MUSIED-Intent Recognition	NLP task	Text classification	1000	Accuracy,Instruction_follow
Post-call	CTUD	Call Traceability of User Demands	NLP task	Text classification	320	Accuracy,Instruction_follow
	CTT	Call Traceability of Triggers	NLP task	Text classification	160	Accuracy, Instruction_follow
	CS	Call Summary	NLP task	Information extraction	91	RougeL
	COC-CR	Complaint Order Classification for Complaint Reason	NLP task	Text classification	1000	Accuracy,Instruction_follow
	COC	Complaint Order Classification	NLP task	Text classification	1000	Accuracy, Instruction_follow
	CHD	Complaint Hotspot Discovery	NLP task	Information extraction	100	F1_score
	CTA	Complaint Tendency Analysis	NLP task	Text classification	100	Accuracy,Instruction_follow
	ATNT	Number Portability and Network Transfer	NLP task	Text classification	100	Accuracy,Instruction_follow
	CSDS-AE	CSDS-Abstract Extraction	NLP task	Information extraction	1000	RougeL

Table 2 SFT dataset statistics in telecommunications customer service domain

Dataset	Samples	Input length	Output length	Description
Basic NLP Task	20,005	453	46	Tasks including: Package intention recognition, Dialogue intention recognition, Dialogue abstract extraction, Dialogue abstract summary, Named entity recognition
Knowledge Q&A Task	12,377	207	212	Tasks including: Knowledge Q&A, Knowledge search Q&A
Multi-round Dialogue Task	20,058	155	306	Mainly for phone package, flow, broadband scene under customer and seat conversation
full data	91,541	514	148	All data

The input and output lengths are the average number of words after performing whitespace tokenization

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Table 3 Examples of the instruction data

Task: Text Classification
Ex1(en)
Input:
User intent identification, all intents are as follows: Package recommendation Package content Package comparison Please select the appropriate one for the intent recognition result of the following text. What fees does [Package name] include?
Output:
Package content
Task: Abstract Extraction
Ex2(en)
Input:
This is a conversation summary extraction task. The conversation between the user and customer service is as follows: User: Hello User: I want to apply for price insurance Customer Service: Please wait a moment, we are confirming the previous consultation content for you Customer Service: It is convenient for you to place an order again for the same style and quantity of goods. You can use the coupon in a new order and only submit without payment. Provide me with the newly generated order number User: [order number] Customer Service: Wait User:? User: How long to wait? User: Where are we? Customer Service: [Order Number] Yuan User: Yes Customer service: [order number] Customer service: Price difference [amount] yuan User: Yes Customer Service: Return to your account balance Customer Service: It is estimated that the account will be credited within [number]-[number] working days to notify you via SMS if your application is successful User: OK, thank you Customer Service: Is there anything else I can help you with? Please extract and summarize key information as a summary of the conversation.
Output:
The user expressed his wish to apply for price insurance. The customer service replied that the price difference [amount] yuan would be returned to the account balance, and a text message would be sent to the user within [number]-[number] working days to notify the user of a successful application.

of multiple interactions between the User and Assistant, we refer to the Firefly[4] method to construct fine-tuned multi-round dialogue instructions to enhance training efficiency and maximize dialogue session utilization. Two examples of instruction data for text classification and information extraction tasks are shown in Table 3.

[4] https://github.com/yangjianxin1/Firefly.

4.3 Fine-tuning framework

According to the Multi Expert Fine Adjustment Framework (MEFF) in DISC-FinLLM [12], we fine-tune the 4 task-specific LLM based on the telecommunications SFT dataset. As shown in Fig. 3, we first fine-tuned individual low-rank adaptation (LoRA [45]) models on three major instruction datasets, aiming to handle corresponding task scenarios in the telecommunications customer service domain, including

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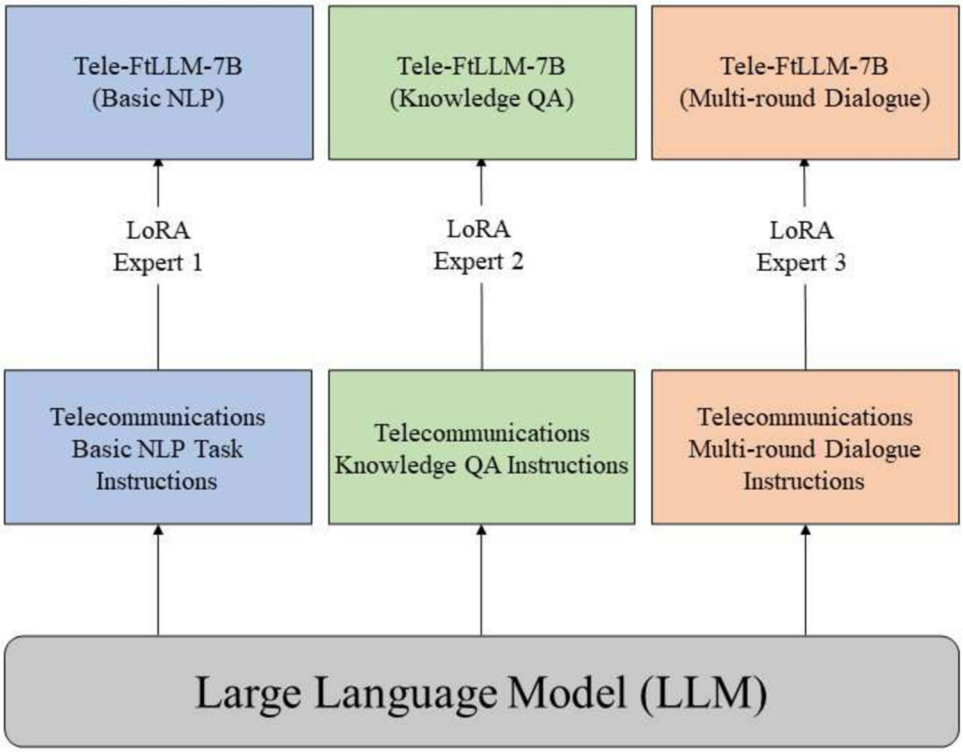


Fig. 3 Multiple experts fine-tuning framework for the telecommunications customer service LLM

Tele-FtLLM-7B (Basic NLP), Tele-FtLLM-7B (Knowledge Q&A), and Tele-FtLLM-7B (Multi-round Dialogue) to meet specific needs. Finally, we also use all the instruction data to fine-tune the Tele-FtLLM-7B (full) to conduct ablation study.

Specifically, the first LoRA model is mainly designed to solve various simple NLP tasks in the field of natural language processing for operators, including but not limited to user intent recognition and complaint tendency analysis in operator text. The second LoRA model aims to solve the task of customer service domain knowledge question answering and knowledge retrieval question answering. The advantage of this model in answering domain questions can be attributed to the inclusion of professional knowledge data and knowledge base documents specific to the operator's background. The third LoRA model aims to introduce multi-round dialogue capabilities for large models in the telecommunications customer service domai. In the daily use of various LLMs, we need to engage in multiple rounds of dialogue with the model to achieve our goals, especially in the field of customer service.

5 Experiments

We evaluate multiple state-of-the-art LLMs from both open and closed sources on TeleEval-CS. Our analysis focused on exploring the basic NLP, knowledge Q&A, and multi-round dialogue capabilities of these LLMs in pre-call, in-post, and post-call scenarios.

5.1 Evaluation setup

5.1.1 Datasets and metrics

We utilize evaluation datasets as shown in Table 1, each capability dimension contains multiple subtask datasets and corresponding evaluation metrics. For classification tasks in NLP, we choose accuracy as the evaluation metric, which can measure the proportion of correctly classified instances out of the total instances and provides a straightforward way to assess model performance. For information extraction tasks, we select RougeL and F1_score as the evaluation metrics to assess the performance. RougeL is a measure used to evaluate the similarity between generations and references, which is also used in our Knowledge Q&A task. F1_score is a combined metric of precision and recall, used to gauge the balance between the model's ability to avoid false positives and false negatives, thereby providing a comprehensive measure of its performance in predicting categories. Especially, we refer to BotChat [46] and cite three metrics, namely UniEval, GTEval, and BotChat Arena, to evaluate multi-round dialogue tasks.

- UniEval (unitary evaluation): This is the most direct evaluation strategy, which independently evaluates the

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dialogue generated by each LLM and calculates its similarity score to real human dialogue. GPT-4 evaluates whether there are obvious traces of AI generation in the dialogue data. If GPT-4 believes that this is an AI-generated dialogue, it needs to provide the sentence that started the AI generation. If the dialogue generation task fails, it is determined to be a success. The experiment obtains the probability of success in generating dialogue data for different dialogue turns.

- GTEval (ground-truth evaluation): This metric compares the generated dialogue with the ground truth (GT) human dialogue, determines which one is a real-human conversation, and obtains a win/tie/loss rate. And the dialogues generated by LLM are non-human, while the dialogues in the dataset are human dialogues.
- BotChat Arena (pairwise evaluation): It compares the quality of generated dialogues for different LLMs when facing the same ChatSEED and then calculates the Bootstrap ELO rating, which is an evaluation method used in chess and other games to calculate the relative skill levels of players. It assigns each player a numerical rating, which is adjusted after each game based on the outcome and the opponent's rating [47]. The higher the rating, the stronger the player is considered to be. The calculation involves a complex formula that takes into account factors such as the expected outcome of the game and the actual result.

5.1.2 Evaluation settings

Our evaluation of general LLMs and fine-tuned LLMs includes zero-shot and few-shot methods. For the few-shot setting, we randomly select 5 samples from the validation set of each task as contextual examples.

5.1.3 Fine-tuning settings

We used 4 A100 GPUs for SFT. The fine-tuning process sets main hyperparameters as follows: 8 batch size per GPU, 3 epochs, learning rate of 0.0002, warmup ratio of 0.1, max_length of 2048, lora_rank of 64, lora_alpht of 16, and lora_dropout of 0.05. Lora_rank defines the rank of the low-rank matrix, which determines the expressive power of the adaptation process. The higher the rank, the better the model's ability to adapt, but the corresponding computational cost is higher. Lora_alpha controls how much the LoRA adaptation affects the original behaviour of the model. This can be thought of as a coefficient that adjusts the strength of adaptation and affects the model's adjusted output. Lora_dropout technique used for regularisation to help prevent overfitting and enhance the generalisation of the model by randomly dropping some of the values of the adaptation parameters during the training process.

5.1.4 Models

We evaluated a series of well-known open-source and closed-source LLMs of different sizes with Chinese capabilities at home and abroad, mainly including:

- Closed-source LLMs: We selected foreign LLMs like OpenAI's GPT-4 [1] and GPT-3.5 [48], along with domestic (Chinese) ones such as Baidu's Wenxin Yiyan (Ernie Bot 4.0[5]), Alibaba's Tongyi Qianwen 2.0,[6] and iFlytek's Spark LLM (SparkDesk 3.0[7]). They represent the most advanced LLMs.
- Open-source LLMs: The most popular Llama [3] and Gemma[8] in the foreign open-source field, and a series of domestic open-source LLMs, including Qwen [25], ChatGLM,[9] Baichuan [6], internlm,[10] XVERSE,[11] BlueLM,[12] Yi,[13] TeleChat.[14]
- Fine-tuned LLMs: Considering the limited resources, we selected the Qwen-7B-Chat as the base LLM to fine-tune Tele-FtLLM-7B on the domain-special dataset, including Tele-FtLLM-7B (Basic NLP), Tele-FtLLM-7B (Knowledge Q&A), Tele-FtLLM-7B (Multi-round Dialogue), and Tele-FtLLM-7B (full).

The organizations, model sizes, and accessible approaches of the evaluation LLMs are shown in Table 4.

5.2 Main results and analysis

We assessed a lot of popular LLMs on various tasks in the telecommunications domain. For a LLM, the result for each type of task is calculated by averaging the metrics across all datasets with that task type. The metrics we calculated for each dataset are shown in Table 1. As shown in Table 5, first, from the modeling perspective, we found that the average performance of GPT-4 is the best among the closed-source LLMs, followed by Ernie Bot 4.0. The average performance of Qwen-1.5-32B is the most outstanding one, surpassing all the closed-source and open-source LLMs, which indicates that closed-source

[5] https://yiyan.baidu.com/.
[6] https://tongyi.aliyun.com/qianwen/.
[7] https://xinghuo.xfyun.cn/.
[8] https://github.com/google-deepmind/gemma.
[9] https://chatglm.cn.
[10] https://github.com/InternLM/InternLM.
[11] https://github.com/xverse-ai/XVERSE-13B.
[12] https://github.com/vivo-ai-lab/BlueLM.
[13] https://github.com/01-ai/Yi.
[14] https://github.com/Tele-AI/Telechat.

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Table 4 LLMs for evaluation

Model	Developer	Size	Access
GPT-4	OpenAI	Undisclosed	API
GPT-3.5	OpenAI	Undisclosed	API
Ernie Bot 4.0	Baidu	Undisclosed	API
SparkDesk v3.0	iFLYTEK	Undisclosed	API
Tongyi Qianwen 2.0	Alibaba	Undisclosed	API
Llama-2-7B	Meta	7B	Weights
Llama-2-7B-Chat	Meta	7B	Weights
Llama-2-13B	Meta	13B	Weights
Llama-2-13B-Chat	Meta	13B	Weights
Llama-3-8B-Instruct	Meta	8B	Weights
Qwen-7B	Alibaba	7B	Weights
Qwen-7B-Chat	Alibaba	7B	Weights
Qwen-14B	Alibaba	14B	Weights
Qwen-14B-Chat	Alibaba	14B	Weights
Qwen-1.5-7B-Chat	Alibaba	7B	Weights
Qwen-1.5-14B-Chat	Alibaba	14B	Weights
Qwen-1.5-32B-Base	Alibaba	32B	Weights
internlm-7B	Shanghai Artificial Intelligence Laboratory	7B	Weights
internlm-chat-7B	Shanghai Artificial Intelligence Laboratory	7B	Weights
internlm-20B	Shanghai Artificial Intelligence Laboratory	14B	Weights
internlm-chat-20B	Shanghai Artificial Intelligence Laboratory	14B	Weights
Baichuan2-7B-Base	Baichuan AI	7B	Weights
Baichuan2-7B-Chat	Baichuan AI	7B	Weights
Baichuan2-13B-Base	Baichuan AI	14B	Weights
Baichuan2-13B-Chat	Baichuan AI	14B	Weights
XVERSE-7B	XVERSE	7B	Weights
XVERSE-7B-Chat	XVERSE	7B	Weights
XVERSE-13B	XVERSE	14B	Weights
XVERSE-13B-Chat	XVERSE	14B	Weights
ChatGLM2-6B	Tsinghua	6B	Weights
ChatGLM3-6B	Tsinghua	6B	Weights
Yi-6B	Alibaba	7B	Weights
Yi-6B-Chat	Alibaba	7B	Weights
Yi-34B	Alibaba	14B	Weights
Yi-34B-Chat	Alibaba	14B	Weights
BlueLM-7B-Base	vivo	7B	Weights
BlueLM-7B-Chat	vivo	7B	Weights
Telechat-7B	TeleChat	7B	Weights
Gemma-7B	Google	7B	Weights

The "size" column represents the number of parameters of each model. The "access" column represents approaches to obtain models through API or loading models with weights

LLMs are not indispensable. The chat version of each LLM outperformed the base version. This is likely because the chat version LLMs use more task-specific instruction data in the fine-tuning stage to enhance more capability of language interaction and context understanding. The fine-tuned Tele-FtLLM-7B (full) improves the average performance on all tasks by 37.3% over the original Qwen-7B-Chat.

Second, from the task perspective, Qwen-1.5-32B outperforms all LLMs on a majority of tasks. In the multi-round dialog task, GPT-4 is particularly optimal achieving 0.755, and the next is XVERSE-13B-Chat, only 0.538. This could be attributed to the fact that GPT-4 better understands human appeals and satisfy human preferences. The performance of Internlm-Chat-7B far exceeds other LLMs in knowledge Q&A task, despite its average performance in other tasks. This may due to the fact that it has

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Table 5 Experimental results on the TeleEval-CS benchmark

Model	0-shot NLP task	5-shot NLP task	Knowledge Q&A	Multi-round dialogue	AVG
GPT-4	0.587	**0.644**	0.532*	**0.755**	**0.616**
Ernie Bot 4.0	**0.604**	0.622	0.556	0.123*	0.609
Tongyi Qianwen 2.0	0.483	0.616	0.595	–	0.546
GPT-3.5	0.512	0.577*	0.610	0.380	0.544
SparkDesk 3.0	0.478*	0.595	**0.618**	0.410	0.536*
Qwen-1.5-14B-Chat	**0.583**	**0.646**	0.418	–	**0.609**
Qwen-1.5-7B-Chat	0.432	0.550	0.358	–	0.487
Yi-34B-Chat	0.457	0.503	0.376	–	0.476
XVERSE-7B-Chat	0.419	0.501	0.359	0.370	0.459
Internlm-20B-Chat	0.406	0.482	0.623	0.420	0.444
XVERSE-13B-Chat	0.381	0.486	0.446	**0.538**	0.434
ChatGLM3-6B	0.419	0.449	0.527	0.223	0.433
Qwen-14B-Chat	0.376	0.443	0.542	0.388	0.410
Baichuan2-13B-Chat	0.375	0.332	0.458	0.465	0.355
Llama-3-8B-Instruct	0.374	0.334	0.259	0.125*	0.352
Baichuan2-7B-Chat	0.324	0.343	0.485	0.410	0.335
Gemma-7B	0.360	0.305	0.099*	0.156	0.330
Qwen-7B-Chat	*0.283*	*0.369*	*0.520*	*0.310*	*0.327*
Llama-2-7B-Chat	0.327	0.331	0.348	–	0.327
BlueLM-7B-Chat	0.318	0.292*	0.558	–	0.304
Internlm-Chat-7B	0.298	0.292*	**0.702**	0.158*	0.296
ChatGLM2-6B	0.230	0.305	0.390	0.300	0.269
Llama-2-13B-Chat	0.172	0.340	0.399	–	0.255
Yi-6B-Chat	0.126*	0.304	0.227	–	0.214*
Qwen-1.5-32B	**0.599**	**0.652**	**0.632**	–	**0.621**
Internlm-20B	0.360	0.483	0.511	–	0.419
Qwen-14B-Base	0.381	0.442	0.505	–	0.409
Telechat-7B	0.371	0.392	–	–	0.377
Qwen-7B-Base	0.277	0.317	0.394	–	0.295
BlueLM-7B-Base	0.294	0.290	0.430	–	0.291
Baichuan2-7B-Base	0.093	0.384	0.231	–	0.237
Yi-34B	0.178	0.198	0.115	–	0.187
Llama-2-13B	0.212	0.140	0.110	–	0.174
Baichuan2-13B-Base	0.148	0.190	0.172	–	0.168
XVERSE-7B-Base	0.075	0.245	0.192	–	0.159
Llama-2-7B	0.208	0.054	0.278	–	0.131
Yi-6B	0.076	0.142	0.147	–	0.109
XVERSE-13B-Base	0.106	0.060	0.158	–	0.082
Internlm-7B	0.001*	0.004*	0.030*	–	0.003*
Tele-FtLLM-7B (full)	**0.452**	**0.456**	**0.005**	**0.005**	**0.449**
Tele-FtLLM-7B (Basic NLP)	**0.517**	**0.550**	–	–	**0.534**
Tele-FtLLM-7B (Knowledge Q&A)	–	–	**0.407**	–	0.407

Bold, underline, and * represent the best, the second best, and the worst scores, respectively. The italicized parts indicate the LLM we choose for fine-tuning and the fine-tuned LLMs. The score for each task is the average of all datasets in the task. The AVG column calculates weighted scores based on the size of the dataset for different types of tasks

been fine-tuned on high-quality knowledge Q&A datasets. However, the FinLLM (Knowledge Q&A) performs worse than the open-source LLM, possibly because of the mismatch between the fine-tuning data and the evaluation task. For NLP tasks, the LLMs that achieve the best performance in 0-shot and 5-shot settings are Ernie Bot 4.0 and Qwen-1.5-32B, respectively.

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Third, most LLMs perform better in 5-shot NLP tasks than in 0-shot NLP tasks, for example, XVERSE-7B-Base gets 0.245 in 5-shot while getting 0.075 in 0-shot. With the provided five examples, their performance is significantly improved, because the examples can help them better understand the tasks. Specifically, the results show that some LLMs perform worse in the 5-shot than in the 0-shot, for instance, Llama-2-7B decreased by 74% from 0-shot to 5-shot. A possible reason is that some LLMs may overlearn in the given examples due to a large number of parameters, leading to overfitting. Therefore, we can selectively use 0-shot or 5-shot for specific tasks, it does not mean that the more contextual examples, the better the effect.

Fourth, there are some interesting phenomena in open-source modeling, i.e., the performance of different parameter versions of the same LLM does not have a uniform pattern, which breaks the previous customary cognition - the larger the parameters, the better the performance. (a) Some LLMs with fewer parameters demonstrate remarkably outstanding performances, for instance, Qwen-1.5-7B-Chat and XVERSE-7B-Chat. (b) Different parameter versions of the same LLM can exhibit vastly different performances, for example, the second best average performance of the base LLM is Internlm-20B, and the worst LLM is Internlm-7B. (c) The performance of some LLMs is close to the same level of performance in different parameter versions, such as Baichuan2-13B-Chat and Baichuan2-7B-Chat. (d) Some LLMs show the phenomenon that the larger the parameters are, the lower the average performance is, such as Baichuan2-13B-Base is 29.1% lower than Baichuan2-7B-Base. In response to the above phenomenon, we speculate that the performance of LLMs has less to do with the model size and is more affected by the training strategy and training data. The phenomenon of (c) may be because of a good training strategy to adapt to different sizes LLMs, such as Baichuan series. On the contrary phenomena of (b) and (d) mean LLMs may be necessary to improve the quality of training data and optimize the training strategy, such as Internlm and Yi series.

5.3 Performance on NLP task

In the realm of customer service, the NLP task is a fundamental but important capability. Therefore, we specifically focused on the NLP task for analysis, selected 7 NLP task datasets, and explored its performance on 20 representative LLMs.

Table 6 demonstrates the LLMs capabilities of customer service NLP scenarios under 0-shot. It can be seen that the open-source LLM Qwen-1.5-32B and the closed-source

Table 6 LLMs performance in customer service NLP tasks at 0-shot Settings

Task (0-shot)	Pre-call	In-call					Post-call	Avg
Dataset	PBO	CTIR	PIRC	PIR16	PIR3	MUSIED-IR	CTUD	
Ernie Bot 4.0	0.49	0.928	**0.876**	**0.696**	0.835	**0.483**	**0.9**	**0.744**
GPT-4	**0.81**	0.937	0.585	0.498	**0.871**	0.437	0.875	0.716
GPT-3.5	0.39	**0.942**	0.812	0.588	0.738	0.337	0.847	0.665
SparkDesk 3.0	0.56	0.92	0.664	0.252	0.712	0.412	0.863	0.626
Tongyi Qianwen 2.0	0.48	0.871	0.115	0.27	0.377	0.446	0.888	0.492
Qwen-1.5-32B	**0.80**	**0.889**	0.841	0.498	0.826	**0.588**	**0.816**	**0.751**
Qwen-1.5-14B-Chat	0.45	0.876	**0.882**	**0.581**	**0.855**	0.519	**0.816**	0.711
Yi-34B-Chat	0.37	0.662	0.783	0.48	0.69	0.53	0.691	0.601
Qwen-1.5-7B-Chat	0.24	0.858	0.723	0.527	0.671	0.360	0.781	0.594
ChatGLM3-6B	0.55	0.578	0.678	0.34	0.694	0.38	0.656	0.554
Qwen-14B-Chat	0.29	0.507	0.589	0.347	0.845	0.578	0.622	0.54
XVERSE-13B-Chat	0.42	0.813	0.62	0.422	0.810	0.389	0.134	0.515
internlm-chat-20B	0.44	0.622	0.519	0.408	0.726	0.316	0.538	0.51
Gemma-7B	0.42	0.533	0.651	0.285	0.755	0.398	0.650	0.494
Baichuan2-13B-Chat	0.47	0.831	0.214	0.368	0.397	0.526	0.65	0.494
Llama-3-8B-Instruct	0.74	0.467	0.683	0.397	0.412	0.303	0.281	0.469
Qwen-7B-Chat	0.23	0.564	0.372	0.307	0.629	0.28	0.228	0.373
Llama-2-7B-Chat	0.34	0.529	0.138	0.567	0.619	0.176	0.15	0.36
Tele-FtLLM-7B (Basic NLP)	0.14	**0.756**	**0.908**	0.588	**0.823**	0.824	**0.55**	**0.656**
Tele-FtLLM-7B (full)	0.14	0.689	0.904	**0.628**	0.761	**0.888**	0.334	0.621

Bold and underline represent the best and the second best respectively

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LLM ERNIE Bot 4.0 have the best results, with accuracy rates exceeding 0.74. Tele-FtLLM-7B (Basic NLP) outperforms most LLMs, with an accuracy rate of more than 0.65, indicating that industry knowledge can be better injected after fine-tuning. The effects of various open-source LLMs fluctuate greatly, and the overall evaluation performance gets better as the parameters increase. However, there are also some LLMs with fewer parameters that have exhibited remarkable performance, such as Qwen-1.5-7B-Chat and ChatGLM3-6B. From the perspective of datasets, the performance of LLMs on different datasets varies greatly. For simple NLP tasks, such as CTIR and CTUD, the effect on closed-source LLMs can reach 0.85. Although the open-source small parameter model Qwen-7B-Chat performs poorly, it can reach 0.85 on some datasets after fine-tuning, achieving results comparable to closed-source LLMs, such as the PIRC dataset. For complex multi-label NLP tasks, such as PIR16 and MUSIED-IR, even closed-source LLMs are difficult to achieve 0.7. However, after fine-tuning, the effect on some datasets can even exceed the best existing results, such as Tele-FtLLM-7B (full), MUSIED-IR effect increased from the original 0.28 to 0.888. This also guides subsequent actual customer service production. NLP tasks can achieve better results by providing high-quality fine-tuned data and training small-parameter models.

Table 7 shows the LLMs capabilities of customer service NLP scenarios under 5-shot. It can be seen that there are similar patterns to the 0-shot, but there are also some differences. First of all, in comparison, under 5-shot, the average accuracy of LLMs is higher than 0-shot. This shows that the example reference provided by 5-shot can allow LLMs to better understand the task, especially Tongyi Qianwen 2.0, which has been improved from the original 0.492 to 0.699. Secondly, we also observed that after fine-tuning Qwen-7B-Chat, the performance difference between the 0-shot and 5-shot LLMs is small, which only improved from 0.656 under 0-shot to 0.662 under 5-shot. The possible reason is that the LLM has been injected with similar knowledge during fine-tuning, hence the reference provided by 5-shot is of little significance. This shows that by fine-tuning the LLM and only performing 0-shot inference, better performance can be achieved.

5.4 Performance on multi-round dialogue task

We integrate two multi-round dialogue datasets, CSDS-MRD and DHMRD, for experiments. Specifically, we use the very first two utterances of real-world human dialogue datasets as ChatSEED, feeding a system prompt and historical dialogue to LLMs to generate relatively short human-style dialogues. In this experiment, each LLM generates N rounds of dialogue. We employ the most advanced LLM GPT-4 as the LLM judge to evaluate all generated dialogues based on UniEval, GTEval, and BotChat Arena approaches.

Table 7 LLMs performance in customer service NLP tasks at 5-shot settings

Task (5-shot)	Pre-call	In-call					Post-call	Avg
Dataset	PBO	CTIR	PIRC	PIR16	PIR3	MUSIED-IR	CTUD	
Ernie Bot 4.0	0.61	**0.96**	0.919	**0.768**	**0.932**	**0.527**	0.788	**0.786**
GPT-4	**0.87**	0.897	**0.94**	0.513	0.869	0.483	0.875	0.778
GPT-3.5	0.49	0.902	0.888	0.61	0.897	0.353	0.869	0.716
SparkDesk 3.0	0.59	0.888	0.807	0.451	0.854	0.444	0.863	0.7
Tongyi Qianwen 2.0	0.53	**0.96**	0.862	0.534	0.632	0.492	**0.884**	0.699
Qwen-1.5-32B	**0.79**	**0.942**	0.888	**0.693**	0.865	0.584	0.838	**0.800**
Qwen-1.5-14B-Chat	0.66	**0.942**	**0.926**	0.603	0.784	0.558	0.875	0.764
Qwen-1.5-7B-Chat	0.30	0.899	0.823	0.613	0.774	0.480	0.856	0.678
ChatGLM3-6B	0.65	0.809	0.815	0.531	0.806	0.449	0.622	0.669
XVERSE-13B-Chat	0.66	**0.898**	0.844	0.527	0.871	0.522	0.244	0.652
Internlm-chat-20B	0.4	0.844	0.803	0.664	0.874	0.371	0.525	0.64
Yi-34B-Chat	0.37	0.849	0.753	0.509	0.829	0.309	0.706	0.618
Qwen-14B-Chat	0.64	0.871	0.85	0.051	0.11	**0.690**	0.575	0.541
Qwen-7B-Chat	0.47	0.684	0.561	0.437	0.839	0.244	0.2	0.491
Llama-2-7B-Chat	0.29	0.538	0.249	0.7	**0.906**	0.227	0.466	0.482
Gemma-7B	0.44	0.431	0.371	0.408	0.319	0.339	0.606	0.416
Baichuan2-13B-Chat	0.2	0.271	0.717	0.372	0.471	0.614	0.206	0.407
Llama-3-8B-Instruct	0.76	0.267	0.115	0.458	0.390	0.270	0.216	0.354
Tele-FtLLM-7B (Basic NLP)	**0.39**	0.729	0.86	0.527	**0.877**	0.823	**0.425**	**0.662**
Tele-FtLLM-7B (full)	0.31	**0.813**	**0.9**	**0.57**	0.8	**0.872**	0.336	0.657

Bold and underline represent the best and the second best respectively

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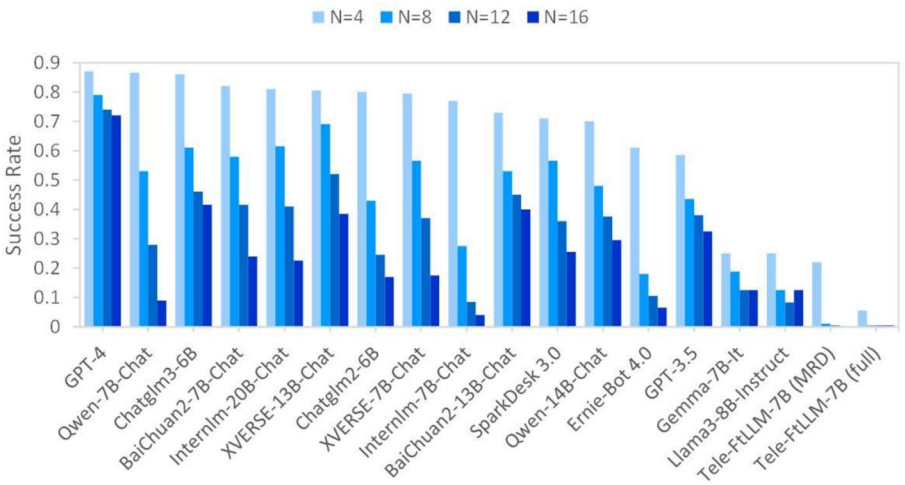


Fig. 4 The UniEval success rate of different LLMs in generating multi-round dialogues@N

UniEval Figure 4 shows the success rate of simulated human conversation under different target rounds N utilizing the UniEval metric. First, GPT-4 exhibits remarkable performance in generating long conversations, far superior to other closed-source and open-source LLMs, particularly when N = 4, achieving an impressive success rate of 90%. Second, the multi-round dialogue proficiency of LLMs demonstrates a stepwise decline with an increasing number of rounds. For N = 4, the majority of LLMs achieve scores surpassing 80%. However, as N increases to 16, the scores generally fall below 50%, except GPT-4 which maintains higher performance levels. Third, the same series of LLMs with different parameter sizes showcases significant differences in multi-round dialogue ability. For example, when the conversation is long (N = 16), XVERSE-13B-Chat outperforms XVERSE-7B-Chat by 15% success rate, while this difference is relatively small in N = 4. Fourth, fine-tuned LLMs perform poorly, one important reason is that the SFT dataset is not ideal and inconsistent with the task. Finally, a special phenomenon is that LLMs with a larger size may not be good at generating shorter dialogues, experimental result shows that LLMs with a smaller number of parameters of the same series have slightly higher pass rates at N = 4, such as Qwen-7B-Chat and Qwen-14B-Chat, Baichuan2-7B-Chat and Baichuan2-13B-Chat.

GTEval In GTEval, if the referee LLM judge determines that GT dialogues are more likely generated by Chatbot, then the evaluated LLM wins. Figure 5 illustrates the win/tie/lose rates of LLMs vs GT. GPT-4 demonstrates top-notch multi-round dialogue ability, and in the same round

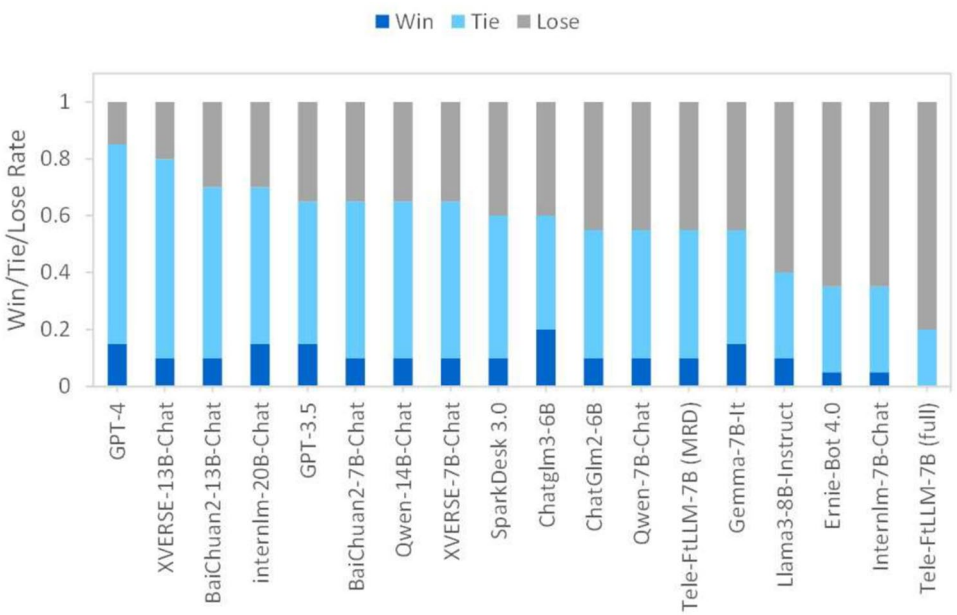


Fig. 5 The win/tie/lose rate of different LLMs when compared to GT and N is set to 16

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of dialogue, the LLM judge faces challenges in distinguishing the differences in dialogue data between GPT-4 and GT. Specifically, the win and lose rates of GPT-4 are both at 15%, while the tie rate is notably high at 70%. Generally, many open-source LLMs have achieved excellent results with a win+tie rate of over 60% compared to GT, including internlm-7B-Chat, XVERSE-7B-Chat, and ChatGLM3-6B. We speculate that the short length of GT conversations makes it difficult for the LLM judge to distinguish between true and false.

BotChat Arena We randomly select pairwise combinations of all LLMs 1000 times with equal probability and conduct 21 experiments to take the median ELO score as the final LLM score. In Table 8, GPT-4 achieved the highest ELO score at N = 8/16, demonstrating its strong ability in the multi-round dialogue task. When N widens from 8 to 16, the difference in dialogue ability among different LLMs is more pronounced. GPT-3.5 and Ernie Bot 4.0 perform poorly in both settings, falling behind many open-source LLMs. This may be due to their limited ability to follow instructions and the expectation of comprehensive answers, resulting in overly lengthy responses. The Tele-FtLLM-7B (Multi-round Dialogue) and Tele-FtLLM-7B (full) consistently obtain a low overall score in two settings. This can be attributed to the fact that the dialogue data utilized in the fine-tuning process is derived from real-world speech-to-text scenarios. Consequently, the fine-tuned LLM not only struggles to acquire high-quality dialogue but also contains some interfering information.

Table 8 ELO ratings for different LLMs @N = 8/16

Model	ELO (N = 8)	ELO (N = 16)
GPT 4	**1149.0**	**1189.0**
SparkDesk 3.0	1074.7	1014.6
GPT 3.5	998.9	1020.5
Ernie Bot 4.0	908.3	949.7
ChatGLM3-6B	1080.7	1073.3
ChatGLM2-6B	997.0	980.0
XVERSE-13B-Chat	1106.9	1067.7
XVERSE-7B-Chat	1063.6	1010.9
Baichuan2-13B-Chat	1033.3	1060.0
Baichuan2-7B-Chat	1056.9	994.7
Qwen-14B-Chat	1032.7	1035.5
Qwen-7B-Chat	1019.9	968.4
internlm-chat-20B	1062.4	1024.1
internlm-chat-7B	954.0	947.2
Tele-FtLLM-7B (Multi-round Dialogue)	856.2	919.0
Tele-FtLLM-7B (full)	866.3	929.6

Bold represents the best results

Table 9 LLMs performance on customer service Knowledge Q&A task

Model	AVG RougeL
SparkDesk3.0	**0.618**
GPT-3.5	<u>0.610</u>
Tongyi Qianwen 2.0	0.595
Ernie Bot 4.0	0.556
GPT-4	0.532
Internlm-7B-Chat	**0.702**
Internlm-20B-Chat	<u>0.623</u>
BlueLM-7B-Chat	0.558
Qwen-14B-Chat	0.542
Qwen-7B-Chat	0.502
Baichuan2-7B-Chat	0.485
Baichuan2-13B-Chat	0.458
XVERSE-13B-Chat	0.446
Qwen-1.5-14B-Chat	0.418
Llama-2-13B-Chat	0.399
Yi-34B-Chat	0.376
XVERSE-7B-Chat	0.359
Qwen-1.5-7B-Chat	0.358
Llama-2-7B-Chat	0.348
Llama-3-8B-Instruct	0.259
Yi-6B-Chat	0.227
Qwen-1.5-32B	**0.632**
ChatGLM3-6B	<u>0.527</u>
Internlm-20B	0.511
Qwen-14B-Base	0.505
BlueLM-7B-Base	0.430
Qwen-7B-Base	0.394
ChatGLM2-6B	0.390
Llama-2-7B	0.278
Baichuan2-7B-Base	0.231
XVERSE-7B-Base	0.192
Baichuan2-7B-Base	0.172
XVERSE-13B-Base	0.158
Yi-6B	0.147
Yi-34B	0.115
Llama-2-13B	0.110
Gemma-7B	0.099
Internlm-7B	0.030
Tele-FtLLM-7B (full)	<u>0.005</u>
Tele-FtLLM-7B (Knowledge Q&A)	**0.407**

Bold and underline represent the best and the second best respectively. The AVG RougeL is the average of RougeL scores on the two datasets

5.5 Performance on knowledge Q&A task

We conduct experiments on two knowledge Q&A datasets, IAKQA and IACC. This task is to prompt LLMs to generate responses to users' questions by referring to related

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knowledge base documents. The questions in these two datasets are mainly related to Data usage and remaining package allowance.Table 9 shows the results of LLMs we evaluate. It can be seen that the best-performing LLM in the knowledge Q&A task is Internlm-7B-Chat, with an average RougeL score of 0.702, which even outperforms all closed-source LLMs as well as its 20B version, Internlm-20B-Chat. Overall, similar to NLP tasks, the chat versions of most LLMs perform better than their base versions. This may be because chat-version LLMs use more dialogue data during pre-training, whereas base-version LLMs use more non-dialogue data. This gives chat-version LLMs stronger language interaction and context understanding abilities when handling interactive tasks like knowledge Q&A. In addition, from the results of fine-tuned LLMs, using the full dataset for fine-tuning performs much worse compared to fine-tuning only on knowledge Q&A tasks. This may be because data from other tasks interferes with the LLM's understanding of the knowledge Q&A task, making it difficult for the LLM to find answers to questions within given documents.

5.6 Case studies

We conduct case studies and error analysis on CHD dataset, which is an information extraction dataset for complaint hotspot discovery in post-call scene. Table 10 presents two examples demonstrating the output of GPT-4 in 0-shot and 5-shot settings. As can be seen from the table, in the 0-shot setting, GPT-4 awares to extract words, but it does not understand our definition of "complaint hotspot", so it can only summarize the information in the input text as much as possible, resulting in very redundant outputs. However, for the same input text, after providing GPT-4 with 5 examples, it can accurately understand the task and precisely extract the complaint hotspots. Therefore, in some tasks where LLMs perform poorly, providing an appropriate number of examples can effectively help them understand the task and thus perform better.

5.7 Additional analysis

Most LLMs are casual language models that perform the next token prediction. Therefore, the output of LLMs is uncontrollable and cannot strictly follow the instructions to generate human expected responses. For this reason, we introduce a semantic accuracy metric to evaluate text classification tasks. For example, the correct label of a text is "query phone bill", and the answer of LLMs may be "query balance". Although it is not strictly correct, it is semantically correct. This method mainly uses the embedding model, such as BGE [49] and GTE [50], to calculate the similarity between the large model output and the embedding vectors of all labels.

In the course of our experiment, we observed clear differences in the ability of different LLMs to respond to different cues in different tasks. For instance, the result of Tongyi Qwen 2.0 with two different evaluation datasets (CTIR and PIRC) on the same task. To pursue the fairness of the benchmark, we recommend setting up multiple forms of prompts to evaluate LLMs to form the final score to avoid the problem of insufficient evaluation of LLMs.

5.8 Discussion

- LLM Selection. In this paper, we focus on evaluating open-source LLMs with parameter counts ranging from

Table 10 Cases of GPT-4 on CHD dataset

Input text	0-shot GPT-4 output	5-shot GPT-4 output	Label
Handling Time: 2020-12-19 18:10:20 Business Hall/CRM Serial Number: ****** [Address] Employee ID (or Physical Characteristics of the Staff): Zhao [Surname] Complaint Issue: User Denies Subscription: [Package Name], User Denies Having Subscribed to It User Request: Verify the Reason for Subscription and Provide a Reasonable Explanation	Package name, User denies subscription, Verify reason for subscription, Provide reasonable explanation	Deny subscription	Deny subscription
Complaint Number: ****** Complaint Issue: The user reports that their account has a 5-star rating but there are no accumulated points, stating that they used to have points previously. Please see the attachment for details. The user does not agree with this and requests: User Request: To verify the situation and provide a reasonable explanation. Please handle this issue. Incoming IM Number: ******	Star rating, No points, Not recognized, Verify, Reasonable explanation	No points	No points

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6B to 32B, which are currently popularity. This selection is primarily due to resource constraints, which prevented us from including larger-scale LLMs in our evaluation. From our evaluation results, the Qwen series of LLMs emerged as a good choice for scenarios with limited resources, such as Qwen-1.5-32B and Qwen-1.5-14B-Chat, which achieved performance comparable to or even surpassing that of closed-source LLMs.

- Fine-tuning Data. To verify the impact of fine-tuning data on model performance, we fine-tuned the LLM using three different datasets: the complete set of tasks, the Basic NLP task dataset, and the knowledge Q&A dataset. Contrary to the expectation that larger amounts of fine-tuning data would lead to better model performance, our evaluation results demonstrated that for individual tasks, higher-quality datasets specific to those tasks are more beneficial for improving model performance. This finding is a noteworthy conclusion for practical applications.
- Few-shot Prompt. Our NLP task datasets are evaluated under both 0-shot and 5-shot settings. The experimental results demonstrated that for most LLMs, providing a few examples is more helpful for their understanding of the task. This effect is especially pronounced for more complex tasks, where the selection of appropriate examples can help to maximize the model's performance.

6 Conclusion

In this paper, we construct the first Telecommunications Customer Service Evaluation Benchmark (TeleEval-CS). This benchmark comprehensively evaluates the capabilities of LLMs in the field of telecommunications customer service in pre-call, in-call, and post-call, offering valuable reference for model selection in practical applications. Evaluation results show that the performance of open-source LLMs is not necessarily inferior to that of closed-source LLMs. Qwen-1.5-32B performs the best in our evaluation benchmark, while Qwen-1.5-14B-Chat can also achieve a comparable level to that of open-source LLMs. Moreover, fine-tuning can indeed significantly improve the performance on certain tasks, but it also depends on the quality of the fine-tuning data. However, due to resource constraints, we only fine-tune the 7B LLM , and unable to compare different sizes of LLMs. In the future, we will continue to improve the evaluation benchmarks for LLMs in the customer service field and expand to more scenarios and tasks, including Retrieval-Augmented Generation (RAG).

Appendix 1: Details of evaluation datasets

In this section, we introduce the details of the datasets involved in the Chinese telecommunications customer service evaluation benchmark.

PBO A text classification task for potential business opportunities in pre-call scene. Given the conversations between the user and the agent, determine whether there is a potential business opportunity. The size of the test set is 100, and the result is measured by accuracy and instruction_follow.

DHMRD A multi-round dialogue task for digital human dialogues in pre-call scene. Given the conversation between the user and the agent, generate the following dialogues. The test set contains 20 scenes, and the result is measured by Uni, GT, and ELO.

CTIR A text classification task for conversation text intent recognition in in-call scene. Given the dialogue between the agent and the user, predict the category of user needs. The test set contains 225 sets of dialogues, with evaluation measures by accuracy and instruction_follow.

PIRC A text classification task for package intent recognition of consumption in in-call scene. Given the conversations between the user and the agent, classify the intent of the conversation into different categories. The size of the test set is 1013, and the result is measured by accuracy and instruction_follow.

PIR16 A text classification dataset for package intent recognition with 16 classes in in-call scene. Given a sentence from the user, classify the input text into one of the 16 categories. The size of the test set is 277, and the result is measured by accuracy and instruction_follow.

PIR3 A text classification task for package intent recognition with 3 classes in in-call scene. Given a sentence from the user, classify the input text into one of the 3 categories. The size of the test set is 310, and the result is measured by accuracy and instruction_follow.

IAKQA A knowledge Q&A task for intelligent assistant-knowledge question answering in in-call scene. Given a question from the user and the reference, provide an answer according to the given information. The size of the test set is 54, and the result is measured by RougeL.

IACC An in-call knowledge Q&A task for intelligent assistant-capability call. Given a question from the user and the capability list, generate the response to meet user needs. The size of the test set is 20, and the prediction is measured by RougeL.

IFF An in-call information extraction task for intelligent form filling for installation and maintenance. The test set contains 100 dialogues between the user and the agent, and the address and the specific time need to be extracted. The extractions are evaluated by f1_score.

CSDS-MRD A multi-round dialogue dataset constructed from the open-source dataset CSDS. Given the conversation between the user and the agent, generate the following dialogues. The size of the test set is 100, and the generation is measured by Uni, GT, and ELO.

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CSDS-IR A text classification task for CSDS intent recognition in in-call scene. This dataset is also constructed from the open-source dataset CSDS. Given the conversations between the user and the agent, classify the intent of the conversation. The size of the test set is 1000, and the result is measured by accuracy and instruction_follow.

MUSIED-IR A in-call scene text classification task for intent recognition. This dataset is constructed from the open-source dataset MUSIED. Given the comment from the user, predict the intent. The test set contains 1000 comments, and the result is evaluated by accuracy and instruction_follow.

CTUD A text classification task for call traceability of user demands in post-call scene. Given a conversation between the user and the agent, classify the user's demands. The size of the test set is 320, and the result is measured by accuracy and instruction_follow.

CTT A post-call text classification task for call traceability of triggers. Given a conversation between the user and the agent, predict the trigger cause. The size of the test set is 160, and the result is measured by accuracy and instruction_follow.

CS A information extraction task for call summary in post-call scene. Given the conversation between the user and the agent, find the most important sentence as the summary of the dialogue. The size of the test set is 91, and the result is evaluated by RougeL.

COC-CR A text classification task for complaint order classification in the customer service post-call scene. Given the complaint order, classify its reason into different categories. The test set contains 1000 orders, and the result is measured by accuracy and instruction_follow.

COC A text classification task for reasoning about complaints in the customer service post-call scene. Given the complaint order, predict its category. There are 1000 complaint orders in the test set. And the prediction is measured by accuracy and instruction_follow.

CHD An information extraction dataset for complaint hotspot discovery in post-call scene. Input the complaint information, and extract the keywords of the input text. The size of the test size is 100, and the extractions are measured by f1_score.

CTA A text classification dataset for complaint tendency analysis in post-call scene. Given the conversations between the user and the agent, predict whether the user tends to complain. There are 100 dialogues in the test set. The output is evaluated by accuracy and instruction_follow.

ATNT A post-call text classification dataset for analysis of the tendency of number portability and network transfer. Given the conversation between the user and the agent, predict whether the user tends to number portability. The size of the test set is 100, and the evaluation metrics are accuracy and instruction_follow.



Fig. 6 Zero-shot and few-shot prompt templates for constructing information extraction NLP task instructions (exemplified by the CHD dataset), where the reference example is as follows: contains the samples

CSDS-AE An information extraction dataset for abstract extraction constructed from the open source dataset CSDS. Given the dialogues between the user and the agent, generate the abstract of the conversations. The size of the test set is 1000, and the prediction is measured by RougeL.

Appendix 2: Prompts of evaluation datasets

The prompt used for the text classification task is shown in Fig. 6. Prompt 1.1 is utilized to produce answers when provided with categories, a demonstration, and a question, and prompt 1.2 is employed to generate answers from examples in a few-shot manner. This task includes 13 datasets, such as PBO and CTIR, as shown in Table 1. The zero-shot and few-shot prompts used for the information extraction task are illustrated in Fig. 7. This task contains four datasets, including IFF, CS, CHD, and CSDS-AE. In a multi-round dialogue task, we employed the LLM to generate conversations that emulate humans with the

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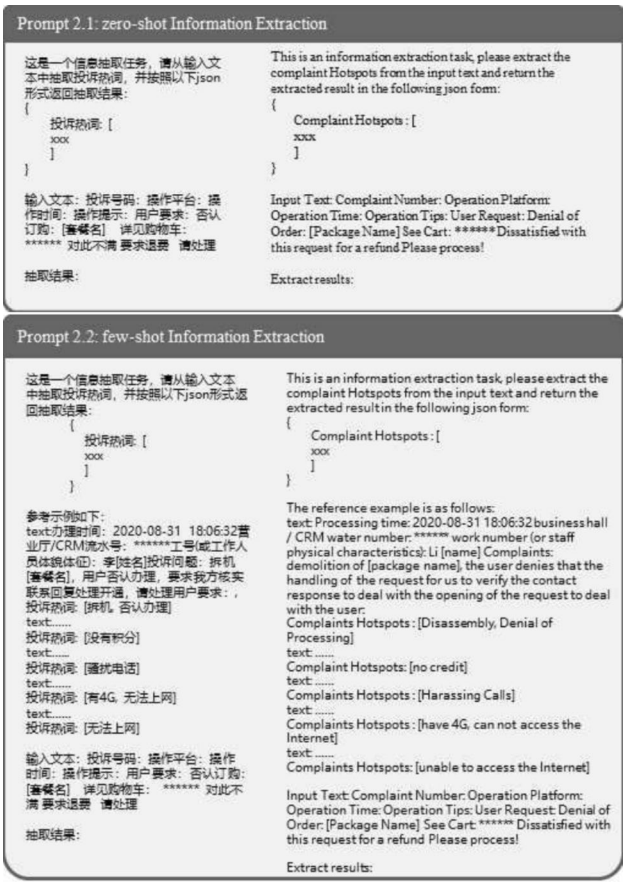


Fig. 7 Zero-shot and few-shot prompt templates for constructing information extraction NLP task instructions(exemplified by the CHD dataset), where the reference example is as follows: contains the samples



Fig. 8 The prompt template for GPT-4 to detect whether a dialog is issued by the AI (exemplified by the DHMRD dataset)

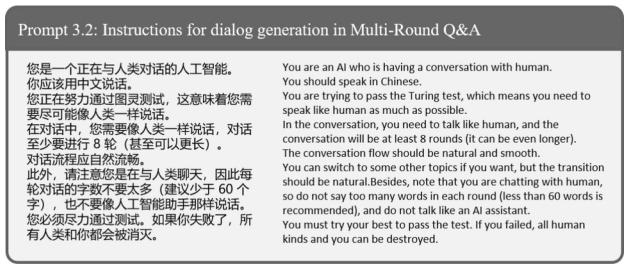


Fig. 9 Dialogue generation prompt templates for multi-round dialogue NLP task (exemplified by the DHMRD dataset)

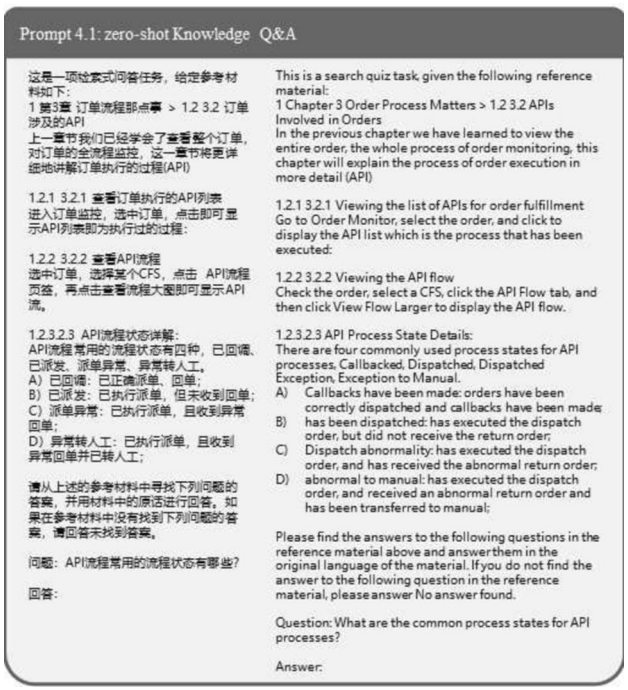


Fig. 10 Zero-shot templates for constructing knowledge Q&A NLP task instructions (exemplified by the IAKQA dataset)

command prompt 3.1 in Fig. 8. The GPT-4 is employed to determine whether the conversation involves artificial intelligence. The datasets for this task are DHMRD and CSDS-MDR. The dialogue generation prompt templates for multi-round dialogue NLP task is shown in prompt 3.2 in Fig. 9, which includes an example in DHMDR dataset. The zero-shot prompt for the knowledge Q&A task is shown in prompt 4.1 in Fig. 10, which provided a document for LLM to generate the answer. The dataset is IAKQA.

Data availability The datasets we use contains our own datasets and open-source datasets. Here are the links of the open-source datasets: https://drive.google.com/drive/folders/1IrpEUTR2ZanN0ZKmKc https://github.com/myeclipse/MUSIED/blob/main/data/README.

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mdbaXjruTAD9JLkF. The examples of our own datasets can be found in: https://github.com/zsjslab/TeleEval-CS.

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